

Dust & Data

What

Dust & Data is a metadata triage tool for museum collections. It samples public artwork records from museum collection APIs, scores the completeness of visitor-facing metadata across five dimensions (image, date, medium, attribution, description), and shows the exact evidence behind every score.

The core job: help a collections team decide which records to investigate first and why. A score flags a metadata review opportunity; it never judges an artwork's quality, its historical importance, or a museum's overall performance.

Current sources are the Cleveland Museum of Art Open Access API and the J. Paul Getty Museum Collection API. The MVP loop: review a sample, inspect and export the actionable records, then press Refresh sample to see a new set. Ordinary app reruns keep the current cached set in place.

Why

Public catalogs are assembled from records created across periods, departments, systems, and cataloging practices, and limited review time makes it hard for staff to choose what to inspect next. A missing-field report says what is empty; an evidence-backed queue also shows the degree of specificity and the source value behind each judgment. That gives catalogers a defensible reason for every flag and produces a CSV that drops into their existing remediation workflow, while aggregate views show managers where problems cluster so work can be assigned and tracked.

For the builder, this is a portfolio prototype demonstrating end-to-end product work in cultural-heritage (GLAM) technology: user-centered problem framing, a transparent scoring policy, and interoperability across two very different museum data models.

Success Criteria

Quantitative:

- A reviewer identifies and validates a small set of actionable records faster than with their current manual process (measured in a walkthrough).
- Share of flagged records judged actionable by a reviewer, with false-positive and false-negative reasons logged.
- Reviewers can explain why a record received its score after viewing Evidence.
- Queue export is used, and the exported fields support the real remediation workflow with no rework.

Technical:

- Records requested, fetched, parsed, skipped, and warned per source and cohort.
- Cold vs cached sample load time; AAT cache hit rate and lookup latency.
- Regression suite passes on the 26-record Getty fixture set and existing Cleveland cases.
- Zero object-ID overlap across consecutive same-size Getty refreshes.

Qualitative:

- In a cataloger walkthrough, participants explain three high-priority cases and challenge any inaccurate gap labels.
- A collections professional reviews the five dimension rules and weights before the score is called decision-ready.

Overview

Background: Cleveland was the first integration because its Open Access API offers a direct REST artwork endpoint with named fields, pagination, filters, and field selection. That made it practical to establish the full product loop: request a cohort, normalize dates, media, attribution, images, and descriptions, score consistently, inspect evidence, and export a review queue. Cohorts of 250, 500, or 1,000 records (default 500) with department-based sampling.

Getty was the deliberate second case: its collection API exposes a graph-oriented Linked.Art JSON-LD model (nested objects, typed statements, stable URIs, SPARQL discovery, IIIF images, vocabulary references). The adapter discovers object URIs through SPARQL, fetches each Linked.Art record, maps fields into a common artwork model, and preserves the original record and URI as evidence. The two sources demonstrate a reusable boundary: each museum owns acquisition and field mapping, while normalization, scoring, queue, charts, and export operate on the common model.

Key terms: a cohort is the bounded record set per review session; the adapter is source-specific fetch-and-map code; AAT is the Getty Art & Architecture Thesaurus used to interpret medium terms; unassessed means the tool could not safely determine a value (its weight is excluded from the score).

Limitation: Getty's 5/25/50/100-object cohorts come from randomly chosen bounded windows of SPARQL object IDs, not a validated representative sample of the collection. Scores are sample-level triage signals, never institution rankings.

Related Links

Internal Documentation: [project README](#), [ARCHITECTURE.md](#), [tests/fixtures/getty/README.md](#) (local repo)

Cleveland Open Access API: <https://openaccess-api.clevelandart.org/>

Getty Collection API docs: <https://data.getty.edu/museum/collection/docs/>

IIIF: <https://iiif.io/get-started/how-iiif-works/>

Collections Trust documentation planning pack:

<https://collectionstrust.org.uk/wp-content/uploads/2018/11/CT-documentation-planning-pack-part-1-2019.pdf>

GitHub: <https://github.com/11samm>

Product Spec

This section gives us a consistent way of making sure we're working on the right problems before diving into solutions.

Alignment

Dust & Data serves two goals. For collections teams, it makes backlog triage repeatable: a defensible first pass over public metadata so limited cataloging time goes to the records with the most actionable gaps, with evidence and CSV export that fit existing remediation workflows. For the builder, it demonstrates end-to-end product work in cultural-heritage technology: problem framing with real users in mind, a transparent and reviewable scoring policy, and interoperability across two institution data models (Cleveland REST, Getty Linked.Art/SPARQL).

Key Insights and Inspiration

- A ranked queue is only useful when each judgment traces to the source field, uncertainty stays visible, and the list is exportable. Evidence is the product; the score is just the ordering.
- The hard part is the scoring rubric for messy real-world data (approximate dates, inconsistent medium text, uncertain attribution), not the code. Defining defensible bands is the product work.
- Missing, withheld, unassessed, ambiguous, and service-failed must never collapse into one result. For Getty images that cannot be verified, the state is unassessed and its 15% weight is excluded from the denominator.
- Vocabulary matching needs role-awareness: the same AAT concept means different things as an object type vs a material vs a technique. The resolution order (embedded relevant AAT ID, then local vocabulary, then cached result, then live AAT search, then unresolved) keeps a plausible match from being presented as a confirmed catalog fact.
- Repeat discovery is the MVP loop: Refresh sample must yield new records, not the same fixed set.
- Inspiration: Collections Trust documentation planning materials (backlog assessment as collections-management work); Getty vocabulary reconciliation guidance; IIF image delivery standards.

Problem Statement

Collections teams know their public catalogs have gaps but cannot see them systematically. Records were created across decades, departments, systems, and cataloging practices: dates range from precise to century-level, medium text uses local synonyms, creators may be named or qualified, descriptions and public images may be absent. With limited review time, staff need a principled way to choose what to inspect next.

How we know this is true: the prototype's own Cleveland and Getty samples surfaced exactly these patterns (approximate dates at risk of reading as exact, medium synonyms needing vocabulary resolution, image-rights vs metadata-rights confusion), and Collections Trust frames backlog assessment and prioritization as core documentation-planning work. Institution schemas also make simple cross-source questions expensive: 'is there a usable image?', 'how precise is the date?', 'does the medium name a material or technique?' Getty's richer Linked.Art model answers more of these but demands role-aware parsing.

Hypothesis & Impact

We believe that giving catalogers a worst-first queue of sampled records with evidence-backed completeness scores will lead to faster identification of actionable metadata gaps versus manual catalog browsing, which will result in a measurable reduction in time-to-first-actionable-record in a walkthrough evaluation.

Guesses for the size of the win: a reviewer working a 25-record Getty cohort names the three most actionable records and explains each score within a 30-minute session; at least half of flagged records are judged actionable by a collections reviewer; the CSV export carries every field needed to enter the museum's existing remediation workflow with no rework. These are targets to measure in user research, not claims.

Constraints

- Public APIs only: no write-back into museum systems; corrections happen in the museum's own tools via CSV export.
- Source availability and rate limits: Getty requires SPARQL discovery plus per-object fetches; vocabulary and IIIF services can fail and must degrade to visible unresolved states, never silent ones.
- Prototype scope: solo builder, Streamlit app, local cache with a 24-hour freshness policy, 26 Getty fixtures for regression (not a production database).
- Unvalidated policy: scoring weights and severity bands are prototype defaults, not an institutional standard; Getty sampling is exploratory, not collection-representative; no cross-museum score comparisons.
- Rights: metadata rights, image rights, and usable-image presence are three separate facts; the UI must never imply an image is cleared for reuse because it can be displayed.

Non-Goals

Full collection audits; correction write-back; curator sign-off workflows; multi-user accounts; automated cataloging; image quality judgments; a full IIIF viewer; a locally hosted copy of all

Getty vocabulary data. Scholarly completeness dimensions (provenance, bibliography, inscriptions, exhibition histories) are out of scope as scoring inputs. The tool triages attention; it does not declare metadata correct.

Design Exploration

This section helps us ensure a collaborative and divergent design process. It also helps us document previously considered directions and rationales for large design decisions.

Initial Ideas

- Core concept: a metadata-health triage dashboard over open museum collection data. This exact framing had not been built on the Cleveland open API; the product framing is the differentiator.
- Hypothetical user stories: as a cataloger, I want a ranked list of records with the biggest public-facing metadata gaps so my review time goes where it matters. As a cataloger, I want the source value behind every gap so I can verify it in the source system. As a manager, I want to see where gaps cluster by dimension and department so I can assign remediation work. As a data steward, I want adapters and fixtures that surface schema drift without silently changing scoring policy.
- Scoring dimensions: image availability (15%), date (25%), medium (20%), attribution (25%), description (15%).

Explored Directions

- Cleveland-first vs Getty-first: chose Cleveland first because its REST API (named fields, pagination, filters) let the product loop be established quickly; Getty second as the interoperability stress test for the adapter boundary.
- AAT handling: considered a live lookup for every medium term; chose the five-step ladder (embedded ID, local vocabulary, cached result, live search, unresolved) to favor source authority, stay deterministic offline, and avoid presenting plausible matches as confirmed facts.
- Image uncertainty: considered penalizing unverifiable Getty images; chose unassessed with the weight excluded from the denominator, since a manifest link is not proof of a usable public image.
- Cohort model: considered one fixed sample; chose the refresh loop (new cohort on demand, cached cohort on reruns) as the MVP discovery behavior.

Key Decisions

- Five scoring dimensions with fixed prototype weights (image 15, date 25, medium 20, attribution 25, description 15); each shown with band, score, weight, and gap explanation.
- Adapter boundary: each museum owns acquisition and field mapping; normalization, scoring, queue, charts, and export operate on the common model.
- Unassessed image state for Getty; rights displayed as three separate facts (metadata rights, image rights, usable image presence).
- Refresh excludes the previous cached cohort; two consecutive same-size Getty samples share zero object IDs (verified live, Sep 2026).
- Scores are sample-level triage signals only: never institution rankings, never cross-museum comparisons.

Decided Direction

Build the Streamlit prototype: source selector (Cleveland with 250/500/1000 cohorts; Getty with 5/25/50/100), overview with score distribution and dimension rates, worst-first queue with per-row gap explanations and a predictable Evidence action, evidence panel (original vs normalized values, AAT resolution path, IIIF preview and manifest with separate rights labels), CSV export of the filtered queue, and the Refresh sample loop with local caching. Chosen because it puts the evidence-backed queue, the actual product bet, in front of a reviewer with the least machinery: no accounts, no write-back, no full-collection crawl.